

REPORT ON PHQ/GAD score distribution analysis

WEEK 2 ----- ANA 2

DATASET USED: GLOBAL MENTAL HEALTH DATASET - 2025

PREPARED BY: DEEPAK BARANWAL (Data Analyst Intern)

PROJECT NAME: AI MENTAL HEALTH ASSISTANT

DATE: 12 – 14 MAY, 2026

1. Introduction:

Mental health disorders such as depression and anxiety are major public health concerns worldwide. This analysis focuses on studying PHQ-9 (Depression) and GAD-7 (Anxiety) score distributions using an open mental health dataset.

The project analyses mental health score distributions across gender and age groups using statistical analysis and visualization techniques such as box plots, violin plots, comparison charts, and trend analysis. The objective of this analysis is to understand PHQ-9 and GAD-7 score patterns, identify demographic trends, and visualize mental health score distributions effectively.

	Age	Gender	Depression_Score	Anxiety_Score	Stress_Level	Sleep_Hours	Mental_Health_History	Treatment	Outcome
0	56	Male	4	10	High	8.2	No	Therapy	Excellent
1	69	Male	22	4	Severe	6.2	Yes	NaN	Fair
2	46	Male	4	2	Low	10.4	Yes	Medication	Good
3	32	Female	19	16	Low	10.0	Yes	Both	Excellent
4	60	Female	14	1	Severe	7.9	No	Therapy	Fair

2. Dataset Description:

The dataset used in this analysis is an open mental health dataset containing demographic and psychological information for multiple individuals.

The following variables were selected for the analysis:

- Age
- Gender
- PHQ-9 (Depression Score)
- GAD-7 (Anxiety Score)
- Stress Level
- Sleep Hours
- Mental Health History
- Treatment Information
- Outcome Status

The dataset contains approximately 2500 records with both numerical and categorical mental health variables.

```
# Dataset information

selected_df.info()

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   2500 non-null   int64
1   Gender                2500 non-null   object
2   Depression_Score      2500 non-null   int64
3   Anxiety_Score         2500 non-null   int64
4   Stress_Level          2500 non-null   object
5   Sleep_Hours           2500 non-null   float64
6   Mental_Health_History 2500 non-null   object
7   Treatment             1892 non-null   object
8   Outcome               2500 non-null   object
dtypes: float64(1), int64(3), object(5)
memory usage: 175.9+ KB
```

3. Data Preprocessing:

Before analysis, several preprocessing steps were performed to clean and structure the dataset. Relevant columns associated with mental health indicators were selected from the original dataset. Missing value analysis was performed to identify incomplete fields. The Treatment column contained missing values, while most other variables were complete. Age groups were additionally created using categorical binning techniques to support age-wise comparative analysis. Python libraries such as pandas and matplotlib were used for preprocessing and visualization.

...	0
Age	0
Gender	0
Depression_Score	0
Anxiety_Score	0
Stress_Level	0
Sleep_Hours	0
Mental_Health_History	0
Treatment	608
Outcome	0
dtype: int64	

4. Statistical Summary:

Descriptive statistical analysis was performed on Age, PHQ-9 scores, GAD-7 scores, and Sleep Hours. The average PHQ-9 (Depression) score was approximately 13.5, while the average GAD-7 (Anxiety) score was around 10.5. Statistical analysis helped identify score distributions, variability, and demographic trends within the dataset.

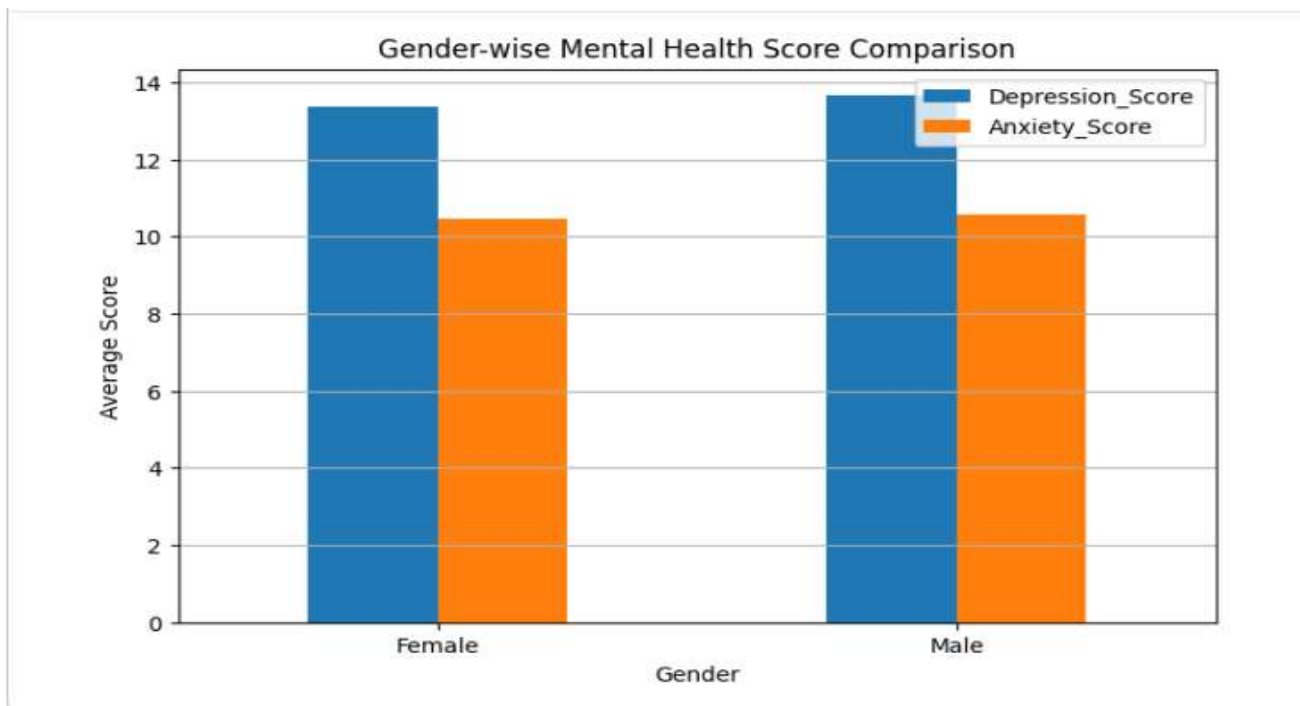
	Age	Depression_Score	Anxiety_Score	Sleep_Hours
count	2500.000000	2500.000000	2500.000000	2500.000000
mean	49.580800	13.518400	10.502800	7.461600
std	18.213933	8.082561	6.329267	2.591345
min	18.000000	0.000000	0.000000	3.000000
25%	34.000000	6.000000	5.000000	5.300000
50%	50.000000	14.000000	11.000000	7.500000
75%	66.000000	21.000000	16.000000	9.700000
max	80.000000	27.000000	21.000000	12.000000

5. Gender-wise PHQ/GAD Analysis:

Gender-wise analysis was performed to compare average PHQ-9 and GAD-7 scores between male and female participants.

The results showed relatively similar mental health score patterns across both genders. Male participants demonstrated slightly higher average PHQ-9 and GAD-7 scores compared to female participants.

Comparative visualization charts were used to better understand gender-based mental health score differences.

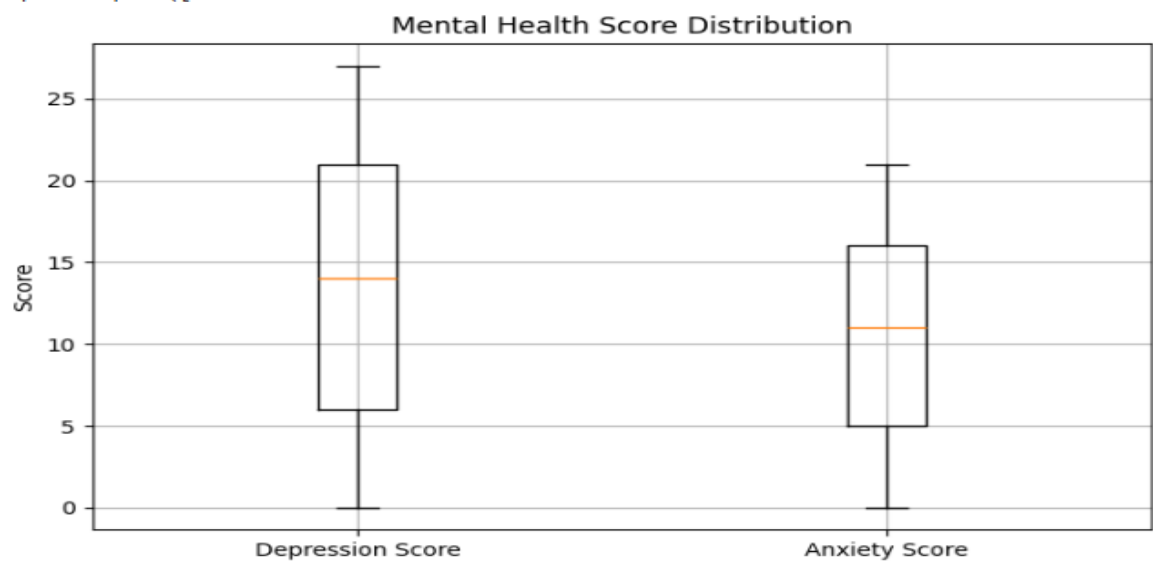


6. PHQ/GAD Distribution Analysis using Box Plot:

Box plot visualization was used to analyse the distribution, spread, and central tendency of PHQ-9 and GAD-7 scores.

The analysis indicated that PHQ-9 scores showed a wider distribution range compared to GAD-7 scores. Median PHQ-9 scores were also comparatively higher.

The box plot visualization helped identify score variability and overall distribution behavior.



7. PHQ/GAD Distribution Analysis using Violin Plot:

Violin plot visualization was used to examine the density and distribution shape of PHQ-9 and GAD-7 scores.

The violin plot highlighted score concentration regions and provided deeper insight into mental health score density patterns compared to traditional box plots.

This visualization improved understanding of PHQ-9 and GAD-7 score distributions within the dataset.

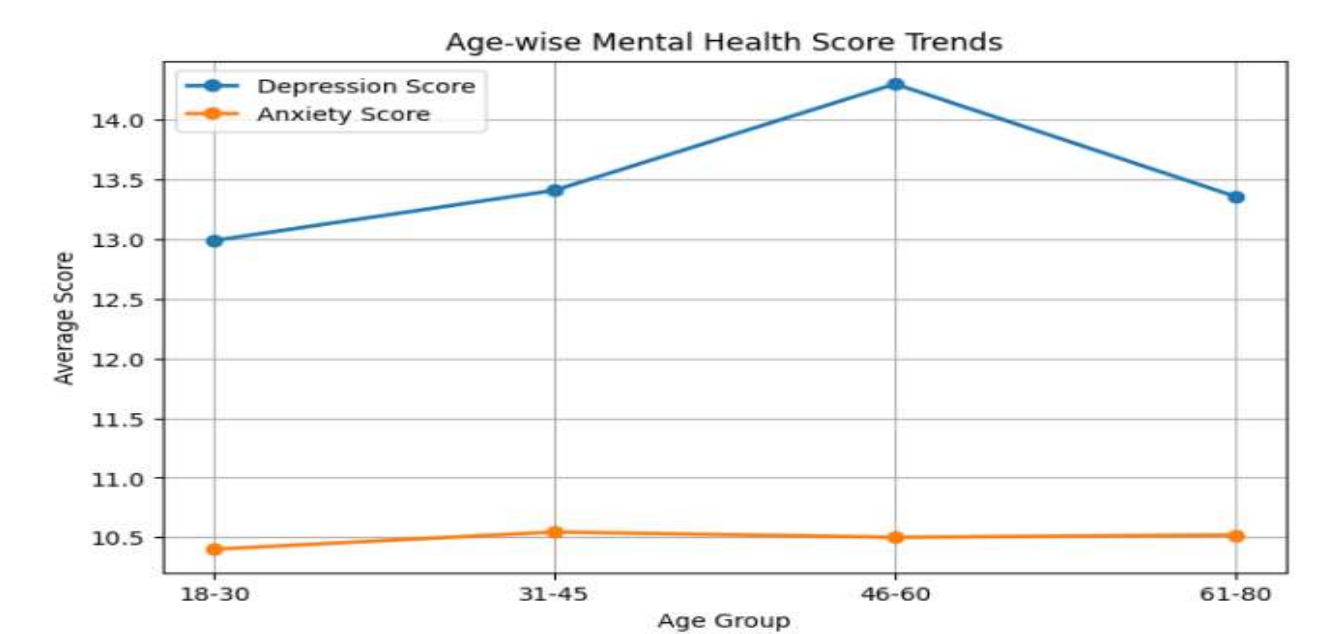


8. Age-wise PHQ/GAD Trend Analysis:

Age groups were created to study variations in PHQ-9 and GAD-7 scores across different age categories.

The analysis revealed that the 46–60 age group showed the highest average PHQ-9 score among all age categories. GAD-7 scores remained comparatively stable across age groups.

Trend line visualizations were used to analyse mental health score patterns across different age ranges.



9. Key Insights:

1. PHQ-9 scores were generally higher than GAD-7 scores across the dataset.
2. Male participants showed slightly higher average PHQ-9 and GAD-7 scores.
3. PHQ-9 score variability was greater than GAD-7 score variability.
4. The 46–60 age group demonstrated the highest PHQ-9 trend.
5. Box plots and violin plots effectively visualized PHQ/GAD score distributions.
6. Age-wise trend analysis helped identify demographic mental health patterns.

10. Conclusion:

This analysis successfully explored PHQ-9 and GAD-7 score distributions using demographic and psychological variables from an open mental health dataset.

The project demonstrated how preprocessing, descriptive statistics, and visualization techniques can be applied to analyse mental health trends across gender and age groups.

Visualization methods such as box plots, violin plots, comparison charts, and trend line graphs provided deeper understanding of PHQ/GAD score behavior and demographic variations.

Overall, the project provided valuable insight into mental health analytics using data science and visualization techniques.